

Abdur Rahman Basil A H

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Summary

Computer Science Engineering student with hands-on experience building full-stack web applications using React, Django, Node.js, Express.js, and MongoDB. Developed and deployed production applications serving 1,000+ students and processing over 215,000 API requests. Experienced in REST API development, authentication systems, database design, Git-based version control, and scalable backend architecture. Passionate about building reliable software solutions that improve user experience and create meaningful social impact.

Education

Saveetha Engineering College, Chennai, Tamil Nadu

B.E. Computer Science and Engineering

CGPA: 8.6

Expected Graduation: 2027

Projects

Trackify

MERN Stack

Live Site — GitHub Repository

- Built and deployed a full-stack academic tracking platform serving **1,000+ students** and processing over **215,000 API requests**.
- Developed **RESTful APIs** for academic credit management, course tracking, and CGPA calculation.
- Designed **MongoDB** schemas and backend architecture to ensure efficient data retrieval and consistency.
- Built responsive and user-friendly **React** interfaces optimized for desktop and mobile devices while improving client-side state management and overall user experience.

Sandboxed Code Execution Engine

FastAPI, Redis, Docker

GitHub Repository

- Built an asynchronous code execution API where submissions are queued via **Redis RQ** and processed by background workers, decoupling job submission from execution and returning a job ID for real-time status polling.
- Designed a **warm-container pool** backed by atomic Redis set operations to eliminate **race conditions** when multiple workers acquire containers concurrently, paired with an autoscaling daemon that maintains a target idle buffer under load.
- Hardened execution containers against untrusted code: disabled networking, capped memory (128MB) and process count, **read-only filesystem** with in-memory scratch space, and a hard 5-second execution timeout.
- Load-tested the system at **100 concurrent submissions** across 5 workers with a **100% success rate**, and identified that execution throughput scales linearly with worker count rather than queue depth.

Club Management System

MERN Stack

GitHub Repository

- Built a full-stack club management platform enabling students and club heads to manage memberships, events, attendance, and academic credits through a **role-based workflow**.
- Implemented secure **JWT authentication** with HTTP-only cookies, **bcrypt** password hashing, and role-based authorization for students and club heads.
- Designed scalable **MongoDB** schemas supporting club memberships, event registration, **OTP-based attendance verification**, notifications, committees, and credit management.
- Developed a complete event lifecycle including registrations, time-conflict validation, OTP attendance verification, attendance approval, and automatic credit allocation using **React, Node.js, Express.js, and MongoDB**.

Skills

Programming Languages	JavaScript, Python, Java, C
Frontend	React.js, HTML5, CSS3, Tailwind CSS
Backend	Django, Node.js, Express.js, FastAPI, REST APIs
Databases	MongoDB, SQL (MySQL, PostgreSQL)
Tools & Platforms	Git, GitHub, Docker, Redis, Postman